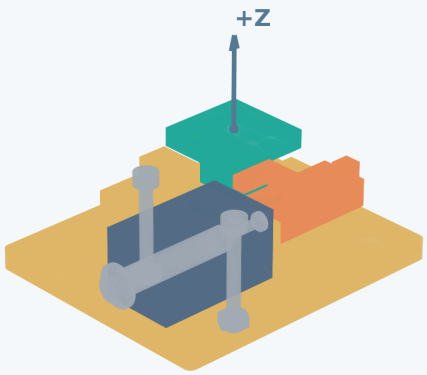
N4 must be pulled along the rail

The concern is correct. The earlier study loaded N4 in +Z, across its T-slot lips. Its adapter departs in +Y. The previous strongest normal-load screen therefore did not establish the strongest extraction lock.

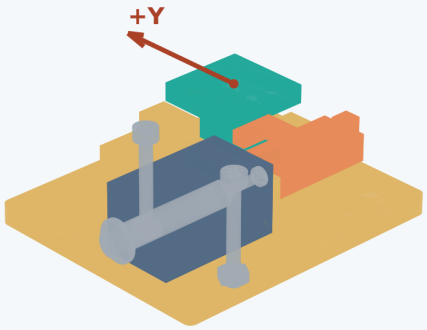
N4 T-RAIL: TWO DIFFERENT LOAD PATHS

Previous test: normal retention



+Z: across the retaining lips

Requested test: locked extraction



+Y: toward the open rail end

Green: payload adapter Orange: keeper Gold: receiver Blue: solenoid

Actual issued CAD; schematic force arrows. Keeper is engaged in both views. Actuator unlocking motion is +X.

New finding: at the common h = 0.55 mm baseline, the three selected extraction-contact cases reach an assumed local PLA threshold at **2.12-2.56 N** (0.48-0.58 lbf). The printed keeper controls, near its lower arm-to-fork junction. This is much lower than the old 24.25 N fully backed normal-rail screen.

These values are finite-mesh, linear-elastic stress-threshold crossings under prescribed seated contacts. They are not physically demonstrated breaking loads, nonlinear collapse loads, safe working loads, or loaded-release limits. Sharp-root stress remains mesh-sensitive.

Scope: unchanged P6 CAD; same P8 PLA; new N4 +Y load path; independent direction audit of all six mechanisms. No physical parts were printed or tested for this addendum.

AUDIT / ALL SIX ADAPTERS

Audit the actual exit direction

Design	Adapter departure	P8 force axis	Finding
N1	+Z normal to receiver	+Z normal	Previous axis matches
N2	+Z normal to receiver	+Z normal	Previous axis matches
N3	+Z normal to receiver	+Z normal	Previous axis matches
N4	+Y along rail	+Z normal	New +Y case required
N5	+Z normal to receiver	+Z normal	Previous axis matches
N6	+Z normal to receiver	+Z normal	Previous axis matches

The solenoid stroke is +X in every model. That motion unlocks the retainer; it is not the payload departure direction. N5 additionally requires its hook to rotate and N6 requires its jaws to open under external load.

Independent rigid CAD checks

The audit reads the issued STEP solids and manifests. A 2 mm attempted motion with each mechanism locked intersects its retaining geometry. With each mechanism fully open, the sampled adapter departure paths are clear. N4 remains blocked under +Z translation even with its keeper withdrawn; the T-foot must slide in +Y before it can leave normally.

The checks are sampled rigid interference calculations. They do not prove automatic opening, loaded actuator motion, print fit, or physical separation. The previous five FEA results are reused because their force axis already matches departure; no redundant rerun is represented as new evidence.

Design	Full backing extraction screen	Two-hole extraction screen
N1 (+Z)	4.24 N	4.24 N
N2 (+Z)	21.50 N	11.26 N
N3 (+Z)	14.49 N	12.74 N
N4 (+Y)	2.47 N	2.47 N
N5 (+Z)	Unresolved	Unresolved
N6 (+Z)	8.41 N	8.41 N

The N4 row uses the full-notch reference branch at h = 0.55 mm; other branches are listed on the results page. N5 contact remains unresolved. N6 assumes equal jaw loading and its two-hole fixture remains mathematical because the rear bore/seat is obstructed. All P8 sharp-root and installation limitations remain. A +Y force is not automatically payload weight.

MODEL / COMPLETE LOAD PATH

A flange pull creates more than pin shear

The applied external pull is P in +Y through the center of the payload mounting face, (30, 12, 17) mm. The lock contact lies near (34.25, 13, 5.55) mm on the adapter and at y = 13.3 mm on the keeper. The 0.3 mm nominal Y gap is taken up before contact.

$F = (0, P, 0)$

$M = r \times F$

Unbalanced moment of flange pull and notch reaction:

$M_x = -11.45\ P$ N·mm; $M_z = -4.25\ P$ N·mm

The adapter therefore needs contact couples at the rail roof/floor and opposite sidewalls. For the full-notch selected branch, the roof and floor contact centroids are 9 mm apart in Y. Their total equal-and-opposite vertical forces are 11.45 P / 9 = 1.27222 P. The opposite X sidewall forces are 4.25 P / 9 = 0.47222 P. These are normal compressive contacts, not friction credit.

Component	Force path used
Adapter	Notch reaction -P in Y; rear rail-roof/front floor couple; opposing sidewall couple; rigid outer flange reaction +P in Y with zero net moment about the stated pull point.
Keeper	Notch applies +P in Y. Local cross-bolt window and remote fork-body guide resist it. Their force/centroid selections close all three moments; broad-guide sensitivity adds a stated roof/floor couple.
Receiver	Equal-and-opposite adapter and keeper contacts pass the complete +Y flange wrench into full rigid underside backing or finite washer-seat fixture areas.
Actuator	No magnetic holding or transverse solenoid support is credited. Neutral rigid keeper motion clears the modeled actuator; real assembly fit still requires measurement.

The adapter submodel fixes its outer flange and applies the retained contact reactions. The resulting flange reaction is the specified extraction force/wrench. This is a cut-boundary equivalent of the flange pull, **not an assembled nonlinear contact solve**. The free keeper uses six scalar rigid-mode gauges whose near-zero reactions are checked; the gauges are not structural supports.

Equilibrium is necessary, but not sufficient

All printed load-path solids use the identical P8 effective PLA: E = 2,200 MPa, Poisson ratio 0.35, tension threshold 25 MPa, compression 40 MPa, maximum shear 15 MPa, density 1.24 g/cm³, assumed 22°C, solid-equivalent/100% infill, short static loading. There is no favorable orientation credit or resolved layer-fracture model. Purchased metal hardware is not reassigned PLA.

Contact case	What changes
Full notch / low guide strips	Uniform pressure over the actual 1.5 × 2.3 mm notch/keeper overlap. The two Y guide-bearing reactions are 0.625 P local and 0.375 P remote, with selected low strips closing the pitch moment.
Upper notch band	Pressure concentrated in an upper notch band, representing one edge-seating sensitivity. Adapter balancing roof/floor couple reduces to 1.2 P; guide reactions are rebalanced.
Broad guide + Z couple	Broader local/remote Y guide pressure; explicitly added compressive roof/floor reactions close the keeper moment. This is a sensitivity, not an established simultaneous contact state.

Clearance and compatibility

The adapter has about 0.3 mm roof clearance, 0.8 mm floor clearance and 0.3 mm side clearance each side. First-order joint rigid seating allows roughly 6.3° pitch and 3.4° yaw while the keeper translates about 0.3 mm in Y. That seating calculation guides contact-family selection but does not solve elastic pressure or finite-rotation contact. Exact STEP interpretation of the small-rotation pose leaves small overlaps (about 0.000616 mm³ at the keeper and 0.001994 mm³ at the receiver).

A separate calculation holding the keeper artificially stationary produced a backstop-amplified force around 3.33 P. It was rejected as the nominal free-keeper condition. It demonstrates sensitivity to constraints; it is not silently used as a multiplier or a measured worst case.

Bambu PLA Basic TDS V3.0 provides context: its reported XY/Z tensile coupon values are 35/31 MPa and moduli 2,580/2,060 MPa. Those 100% infill specimens were annealed/dried. The lower comparison thresholds here are analyst assumptions, not certified material allowables. Source: https://cdn.shopify.com/s/files/1/0645/5876/0155/files/Bambu_PLA_Basic_Technical_Data_Sheet.pdf

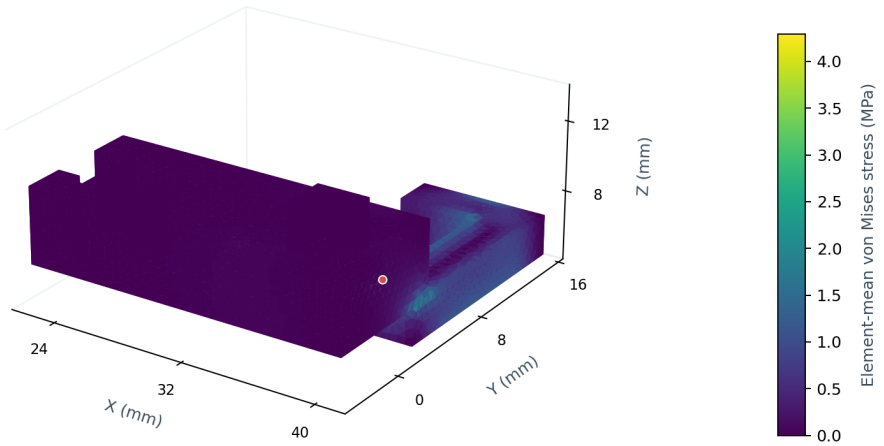
The keeper governs every selected extraction case

Case	Adapter	Keeper	Receiver full / holes	Assembly
Full notch / low guide strips	22.740 N	2.470 N	13.894 / 11.369 N	2.470 N
Upper notch band / guide bearing	18.208 N	2.563 N	14.608 / 11.631 N	2.563 N
Full notch / broad guide with Z couple	22.740 N	2.116 N	13.873 / 11.599 N	2.116 N

The assembly column is the minimum of the component stress screens. Here the keeper is lower than either receiver fixture, so switching fixtures does not change the selected assembly crossing. It does not mean the real mounting joint has been qualified.

N4 keeper: +Y extraction through adapter flange

Actual STEP geometry · 1 N external +Y extraction pull · h=0.55 mm · 22,145 quadratic tetrahedra



Undeformed geometry. Color shows cell-mean stress for visualization, not the PLA failure criterion.
PLA comparison uses principal tension/compression and maximum shear. Red marker projects the raw hot-spot element centroid.
Raw peak tensile 10.12 MPa; compressive 3.707 MPa; shear 4.434 MPa at the unit load.
Prescribed finite contact tractions; ideal mounting. These are simulations, with no physical load rating.

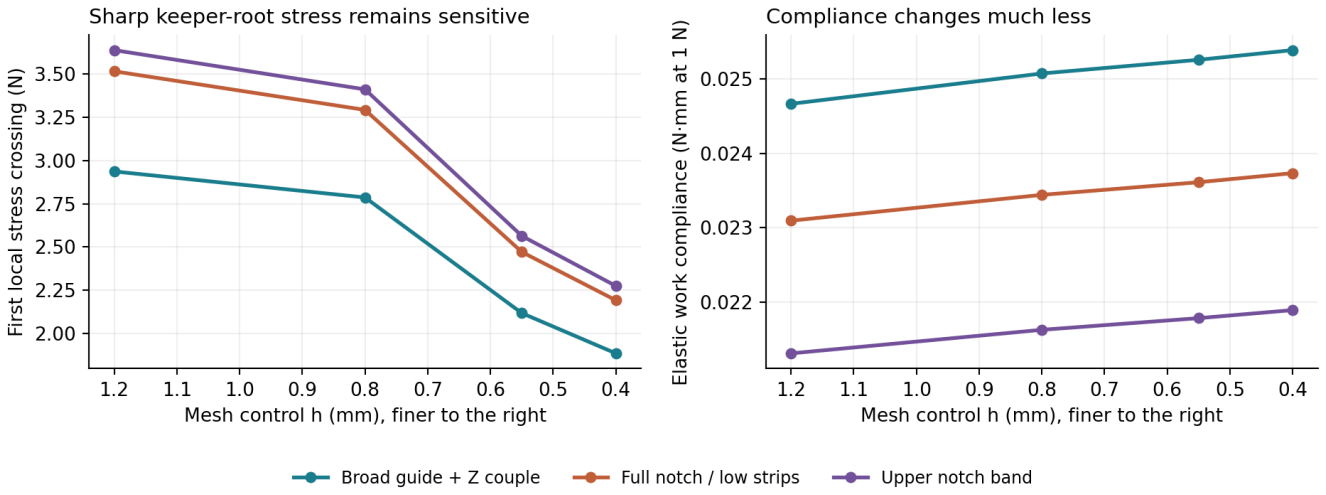
Full-notch case: $\sigma_t = 10.11991\ P$ MPa

$P_{cross} = 25 / 10.11991 = 2.47038$ N

REFINEMENT / RAW LOCAL STRESS

No converged breaking load is claimed

N4 extraction: three contact assumptions, unchanged PLA geometry



Keeper case	h = 1.2 mm	h = 0.8 mm	h = 0.55 mm	h = 0.4 mm
broad guide couple	2.937 N	2.786 N	2.116 N	1.883 N
uniform low edge	3.517 N	3.293 N	2.470 N	2.191 N
upper band	3.639 N	3.412 N	2.563 N	2.274 N

The CAD has an abrupt inner junction between the long return arm and fork body. Raw peak tension increases as this sharp region is refined; the threshold consequently falls. The field remains a mesh-dependent local screen. No arbitrary fillet, excluded hotspot, stress percentile or tensor average is substituted to produce an attractive capacity.

The solver uses quadratic displacement tetrahedra on the actual faceted STEP surface geometry. Baseline h values are 1.2, 0.8 and 0.55 mm, with the additional keeper refinement shown above. h is a mesher control with local refinement boxes, not every tetrahedron edge length. Compliance is much more stable than the raw corner stress.

$$I = \max[\ \max(\sigma_1,0)/25,\ \max(-\sigma_3,0)/40,\ (\sigma_1-\sigma_3)/(2\times 15)\ ]$$

$$P_{cross} = 1 / \max_{domain}(I), \text{ for a 1 N external +Y pull}$$

This criterion locates the first assigned stress threshold in a linear field. It does not predict crack propagation, loss of engagement, ultimate rupture or plastic redistribution. Reported lbf values use 1 lbf = 4.448221615 N.

A strong little tip can feed a weak printed frame

The independent notch calculation gives a contact area A = 1.5 × 2.3 = 3.45 mm². Its uniform mean pressure is p = P/A = 0.289855 P MPa. Reaching the assumed 40 MPa compression threshold by that calculation alone would require 138 N. That number says nothing about bending of the surrounding keeper.

An ideal short cantilever treatment of only the bolt tip uses a 2.4 × 2.3 mm section and 1.05 mm overhang. Its in-plane section modulus is Z = 2.3 × 2.4²/6 = 2.208 mm³, giving σ = P × 1.05/Z = 0.47554 P MPa. The 25 MPa tension threshold would suggest 52.57 N for that ideal isolated section. The complete STEP FEA exposes why that estimate is inadequate: the force also traverses a long offset arm and its sharp junction to the remote guided body.

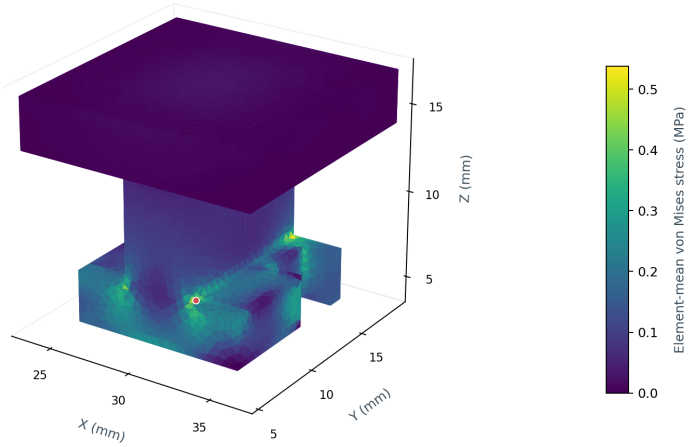
Independent return-arm beam verification

Away from the sharp root, the actual 3 × 2.3 mm return arm has area A = 6.9 mm², I_z = 5.175 mm⁴ and I_x = 3.04175 mm⁴. For the full-notch branch, N = 0.375 P, M_z = -3.75 P N·mm and M_x = -0.43125 P N·mm. These give nominal extreme-section tension 1.304348 P MPa. A fit to 623 fine-mesh cells gives 1.304337 P MPa; the individual axial/bending coefficients agree within 0.081%.

That agreement verifies the force path away from the corner. It does not justify discarding the much higher raw root peak or substituting a nominal-section failure load.

N4 adapter: complete +Y pull and balancing contacts

Actual STEP geometry · 1 N external +Y extraction pull · h=0.55 mm · 30,170 quadratic tetrahedra



Undeformed geometry. Color shows cell-mean stress for visualization, not the PLA failure criterion.
PLA comparison uses principal tension/compression and maximum shear. Red marker projects the raw hot-spot element centroid.
Raw peak tensile 1.099 MPa; compressive 0.9778 MPa; shear 0.4901 MPa at the unit load.
Prescribed finite contact tractions; ideal mounting. These are simulations, with no physical load rating.

The adapter plot includes the notch force and the rail/sidewall moment couples. The receiver receives the complete opposite wrench; it was not replaced with an immovable wall at the bolt tip.

Independent audits check 69 contact-patch uses against the actual STEP faces, pressure direction, force centroids and three-dimensional moment closure. All selected tractions are compressive normal contact loads. Those checks establish geometric/statics consistency, not contact compatibility or material validation.

VERIFICATION / REPRODUCTION

Keep the evidence and the limits together

Check	New study result
Solved scope	36 case/mesh solves; 10 distinct meshes; three actual N4 load-path solids.
Force / moment balance	Max 9.7e-11 N / 3.56e-09 N·mm.
Free residual / gauge reaction	Max 9.91e-10 / 4.11e-12 N.
CAD / mesh volume	Max absolute relative mismatch 0.054%.
Largest model	252,126 displacement DOFs.
Prior verification	The P8 solver was checked against uniform tension and cantilever deflection. The unchanged frozen P9 driver and configuration identities are retained for these new solves.
Physical evidence	No print, coupon, static pull, loaded release or flight test was performed.

Package contents

The analysis ZIP contains unchanged STEP geometry, frozen solver/configurations, all numerical meshes and stress/displacement fields, scalar result CSV, contact and direction audits, independent hand calculations, figure/report source and portable replay instructions. The minimal six-design direction-audit input set is included. P8 comparison scalars are preserved with hashes; the complete P8 field archives remain separate.

Run the documented replay into a new output directory. Do not overwrite the issued results. The report and figures rebuild from the archived fields without rerunning FEA. All displayed numbers come from summary.json/results.csv and the underlying per-case JSON/NPZ files.

PHYSICAL VALIDATION / NOT YET PERFORMED

Pull the locked adapter on its actual exit axis

The numerical and rigid-geometry checks are complete for this correction. The following bench program must still establish the manufactured behavior. Do not use the mesh threshold as a proof load or a safety factor without a defined application and physical evidence.

1. Print matching solid PLA specimens with recorded orientation, settings and measured radii/clearances. Mount the receiver using the intended backing and hardware. Align the load cell with the actual adapter exit axis. For N4, the pull goes through the adapter flange in +Y, with the keeper engaged.

2. Record force and displacement continuously while increasing load. Watch the lower-arm/fork root, guide motion, first crack, permanent set, slip and complete separation. Test at least three specimens per chosen fixture/contact condition for initial screening; establish a statistically justified qualification plan after these exploratory results.

3. Repeat N4 extraction with the intended simultaneous +Z carried load and offset moments. Then command release at measured force, confirming full keeper withdrawal and travel beyond 17.5 mm to clear the rail. A successful coil stroke alone is not a successful separation.

Test condition	What to observe
Locked +Y flange pull	Keeper-root strain/crack, guide motion, unintended slide and fixture slip; record first damage separately from ultimate separation.
Combined carrying and extraction	Apply the actual +Z service load and intended offsets, then increase +Y extraction force. No combined envelope is established by this report.
Commanded unlock and departure	Record actuator stroke and withdrawal force, then independent adapter travel beyond 17.5 mm. Repeat near zero departure force and with realistic misalignment.

Acceptance: no service rating is assigned here. Before qualification, define the actual service wrench and proof factor. A preliminary proof gate requires no crack, no engagement loss and no measurable permanent set beyond a stated fit-sensitive limit; validate that limit against the real mechanism clearance. Any crack, unintended slip or incomplete release fails the selected test condition.